

B R SATHVIK

AI/ML Engineer · Computer Vision · Deep Learning · NLP · RAG Pipelines

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PROFESSIONAL SUMMARY

Motivated AI/ML undergraduate (3rd year) with hands-on experience in computer vision, deep learning, NLP, and model deployment. Engineered and deployed end-to-end AI applications including a medical imaging pipeline with Explainable AI (Grad-CAM), RAG-based LLM tools, real-time object detection systems, and REST API backends. Skilled in data pipeline design, model training, inference optimization, and cloud deployment. Actively seeking AI/ML internship opportunities to contribute to real-world machine learning projects.

TECHNICAL SKILLS

Languages:	Python, JavaScript (Basic)
AI/ML:	PyTorch, TensorFlow (Basic), DenseNet-121, YOLOv8, OpenCV, Scikit-learn, Grad-CAM, HuggingFace Transformers, Model Training, Inference Optimization
NLP & RAG:	LangChain, FAISS, RAG Pipeline, TF-IDF, Cosine Similarity, LLM Integration, Vector Search, Semantic Search, Prompt Engineering
Backend:	Flask, Flask-SocketIO, REST API Design & Development, Streamlit, MVC Architecture
Databases:	SQLite, SQL (Basics), FAISS Vector Database
Tools:	Git, GitHub, Google Colab, Kaggle GPU, Docker (Basic), Power BI, pytest, Agile Workflow
Cloud & AI:	Oracle Cloud Infrastructure (OCI), Groq API, HuggingFace Hub, Streamlit Cloud, Model Deployment

PROJECTS

MediScan AI — Chest X-Ray Disease Detection [\[GitHub\]](#)

Python | PyTorch | DenseNet-121 | Grad-CAM | LangChain | FAISS | LLaMA-3.3-70B | Groq API | Streamlit

- Fine-tuned DenseNet-121 on a 5,863-image NIH chest X-ray dataset; achieved Test AUC-ROC 0.96+ and Test Accuracy 90.2% for NORMAL vs PNEUMONIA classification through rigorous model training and evaluation.
- Engineered Grad-CAM (Gradient-weighted Class Activation Mapping) heatmaps to visualise lung regions influencing model predictions, enabling clinical interpretability and explainable AI output.
- Designed and integrated a Clinical RAG Assistant using LangChain, FAISS vector store, and LLaMA-3.3-70B via Groq API, grounded in WHO pneumonia clinical guidelines — showcasing end-to-end data pipeline and LLM integration.

Edge AI Camera System for Illegal Waste Dumping Detection

Python | YOLOv8 | Flask | Flask-SocketIO | OpenCV | SQLite | REST API | Chart.js | pytest

- Developed and deployed a real-time object detection pipeline using YOLOv8 and OpenCV, integrated with a Flask REST API backend and live analytics web dashboard — achieving real-time inference on video streams.
- Architected a modular MVC-based data pipeline with Flask-SocketIO for real-time alert streaming and SQLite for structured incident logging, ensuring reliable and scalable system design.

AI Knowledge Workspace (RAG + LLM Application) [\[GitHub\]](#)

Python | Streamlit | LangChain | FAISS | HuggingFace Embeddings | Groq API | NLP

- Engineered an AI-powered document interaction platform enabling users to upload PDFs and query content via natural language using a RAG pipeline with semantic search and vector embeddings.
- Implemented document summarisation, semantic search, and AI-powered flashcard generation using LangChain, FAISS vector database, and HuggingFace embedding models — demonstrating full NLP and LLM integration.

Ghost Fishing Gear Detection using YOLOv8 [\[GitHub\]](#)

Python | YOLOv8 | PyTorch | OpenCV | Google Colab | Deep Learning

- Trained a custom YOLOv8 object detection model on a self-annotated marine dataset for abandoned ghost fishing gear detection, achieving 70% validation accuracy.
- Executed the complete ML lifecycle — data collection, annotation, augmentation, model training, inference optimization, evaluation, and documentation — independently.

EDUCATION

B.E. in Artificial Intelligence and Machine Learning

AMC Engineering College, Bengaluru | VTU

CGPA: 7.6

2023 – 2027

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 – AI Foundations	Oracle	2025
Oracle Cloud Infrastructure 2025 – Generative AI	Oracle	2025
HackerRank Certified – Python (Problem Solving)	HackerRank	2024

LANGUAGES & INTERESTS

Languages: English | Telugu | Hindi | Kannada

Interests: Basketball | Volleyball | Skating | Music | Singing